

Synthetic Chemistry II

Organometallic Chemistry

Course Instructors:

Hong Zhang (张弘) zh@xmu.edu.cn

Tingbin Wen (温庭斌) chwtb@xmu.edu.cn



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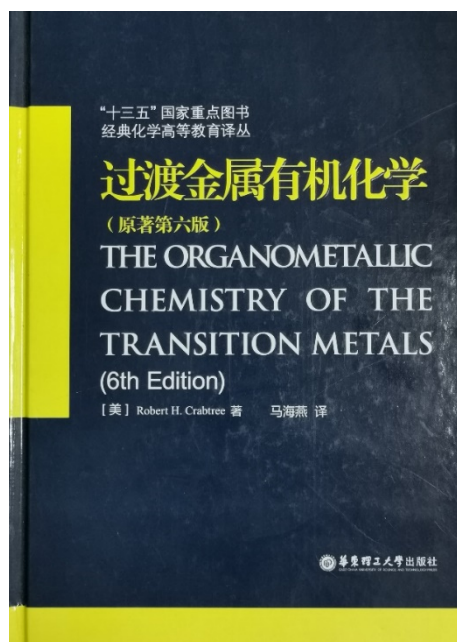




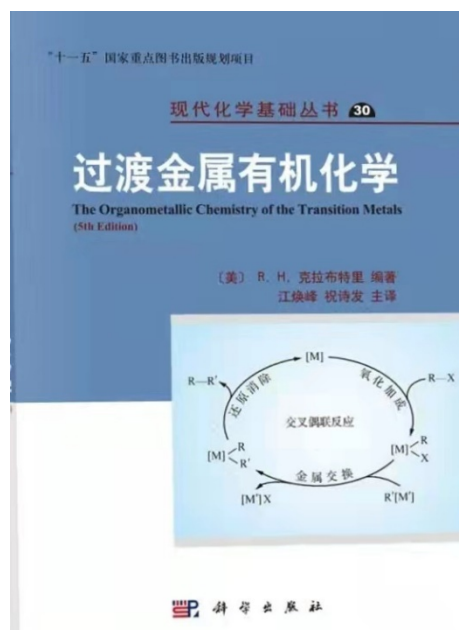
References

本课程不指定教材，以课件作为主要学习材料

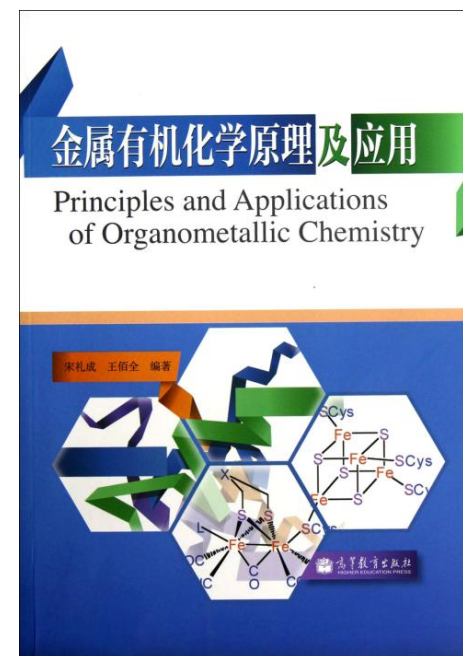
Major Reference Books



《过渡金属有机化学》
(原著第六版)
马海燕 译
2017, 华东理工大学出版社

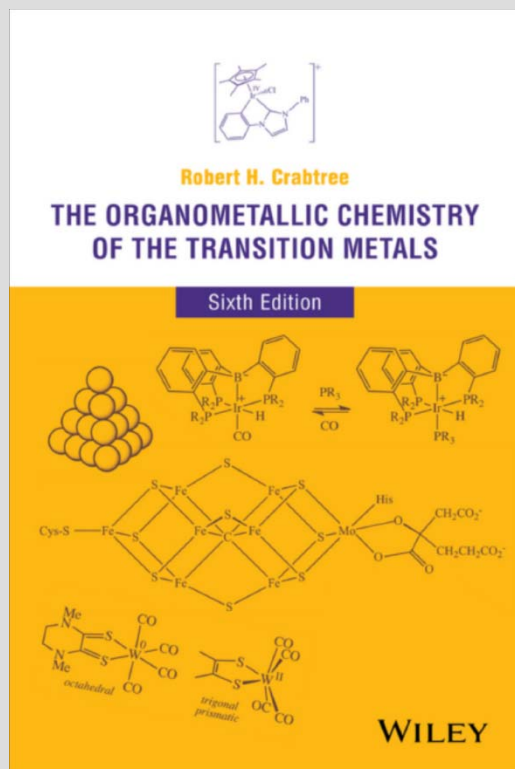


《过渡金属有机化学》
(原著第五版)
江焕峰, 祝诗发 译
2012, 科学出版社



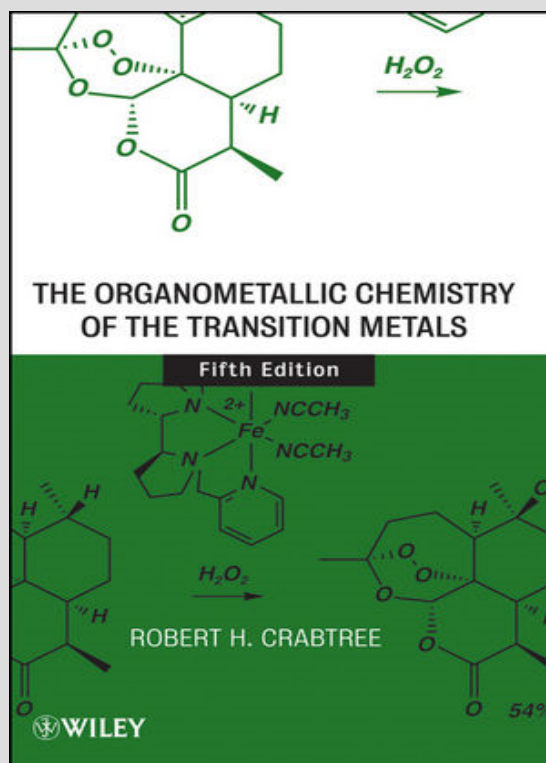
《金属有机化学原理及应用》
宋礼成, 王佰全,
2012, 高等教育出版社

Other Suggested Books



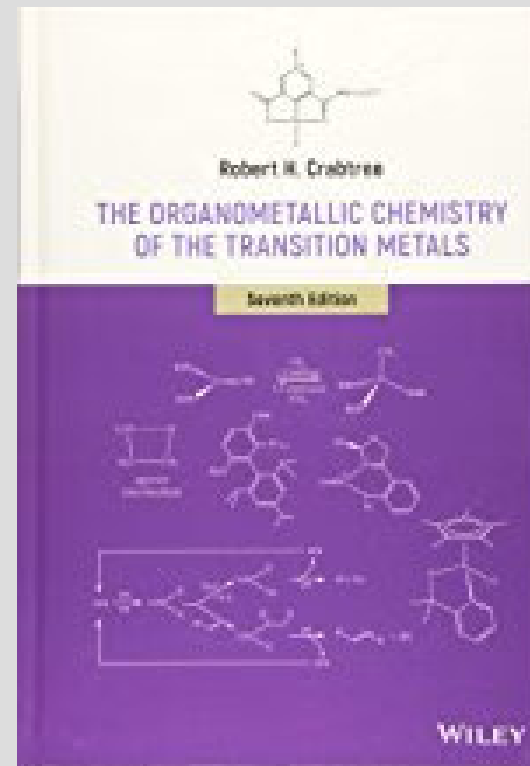
*“The Organometallic Chemistry
of the Transition Metals”*

Robert H. Crabtree
6th Edition, 2014, Wiley



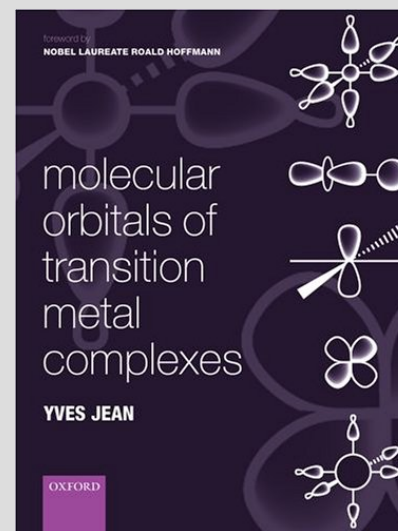
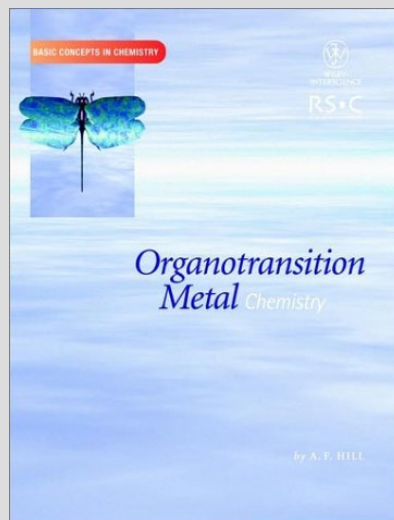
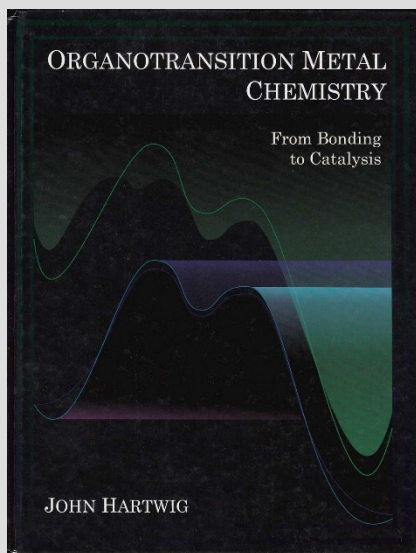
“The Organometallic Chemistry of the Transition Metals”

Robert H. Crabtree
5th Edition, 2009, Wiley
(or 7th Edition, 2019)



Other References:

1. *Organotransition Metal Chemistry: From Bonding to Catalysis*, J. F. Hartwig, University Science Books: Sausalito, 2010.
2. *Organotransition Metal Chemistry*, Akio Yamamoto, 1986.
3. *Molecular Orbitals of Transition Metal Complexes*, Yves Jean, 2005, Oxford University Press.



4. *Organometallic Chemistry*, G. O. Spessard, G. L. G. L. Miessler, Prentice-Hall: New Jersey, 1997.
5. 《金属参与的现代有机合成反应》，麻生明, 广东科技出版社, 2001.
6. 《金属有机化学》何仁, 陶晓春, 张兆国, 华东理工大学出版社, 2007.



Course Outline

PART I

- ✓ Introduction
- ✓ General structure and bonding of organometallic complexes
- ✓ Complexes with metal-carbon σ bond: Metal alkyls, aryl, hydride and related σ -bonded ligands
- ✓ Carbonyls and phosphine complexes
- ✓ Complexes of π -bound ligands

PART II

- Ligand substitution reactions
- Oxidative addition and reductive elimination
- Insertion and elimination
- Nucleophilic and electrophilic addition and abstraction
- Carbene and carbyne complexes
- Homogeneous catalysis

Reaction types in Organometallic Chemistry

—Elementary Reactions in Organometallic Chemistry (基元反应)

- (1) Ligand coordination and substitution
(配体的配位和取代)
- (2) Oxidative addition and reductive elimination
(氧化加成和还原消除)
- (3) Insertion and elimination (de-insertion) reactions
(插入反应和消除(反插入)反应)
- (4) Electrophilic and Nucleophilic Addition and Abstraction
(亲电和亲核加成与攫取)
(Reactions on the complexed ligands 和金属结合的配体的反应)

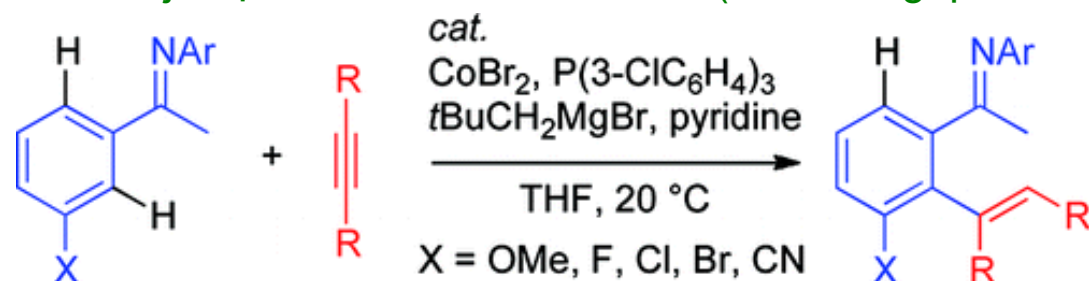
*Help us to understand and predict the **metal-catalyzed or mediated** reactions.*

Be aware:

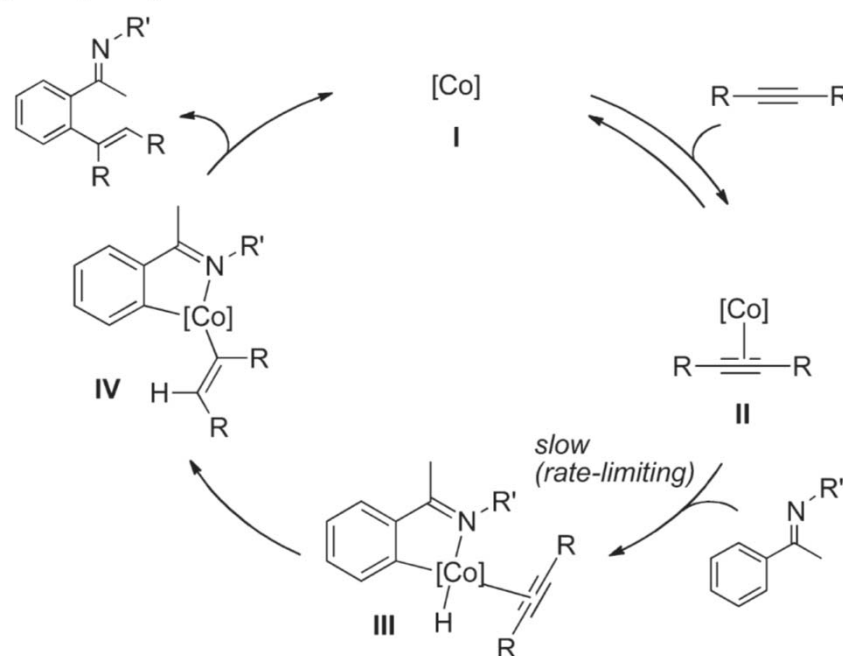
- Classified **in terms of reaction patterns**, not mechanism !!
- An elementary reaction may go through different mechanisms.

Cobalt-Catalyzed, Room-Temperature Addition of Aromatic Imines to Alkynes via Directed C-H Bond Activation

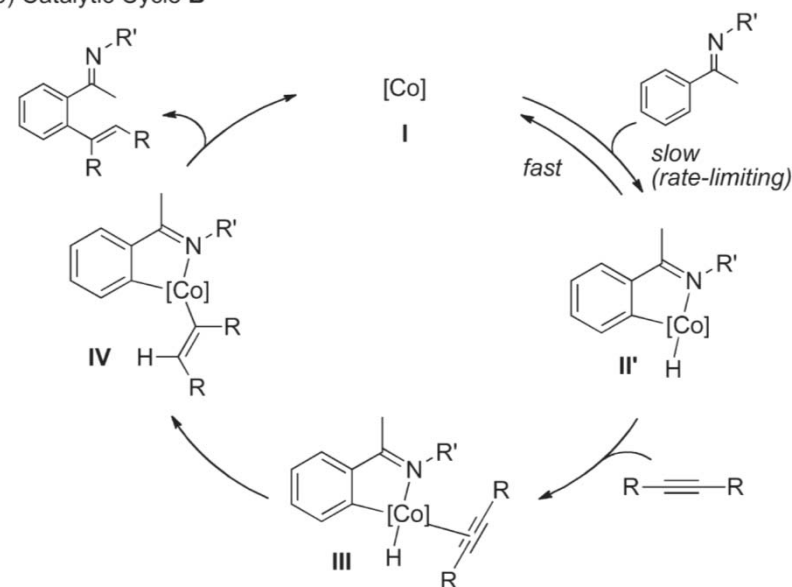
Pin-Sheng Lee, Takeshi Fujita,[†] and Naohiko Yoshikai* (NTU, Singapore 637371, Singapore)



(a) Catalytic Cycle A



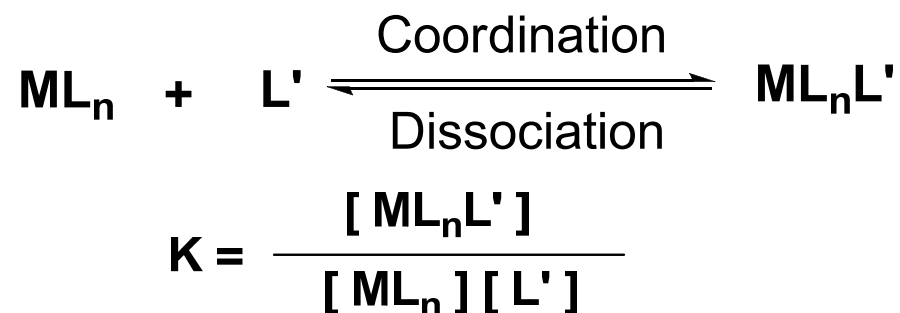
(b) Catalytic Cycle B



A mechanistic investigation into the reaction with the aid of deuterium-labeling experiments and kinetic analysis has indicated that oxidative addition of the ortho C–H bond is the rate-limiting step of the reaction.

Ligand Coordination and Substitution

The first step in homogeneous catalysis (**Metal Coordination Catalysis**)

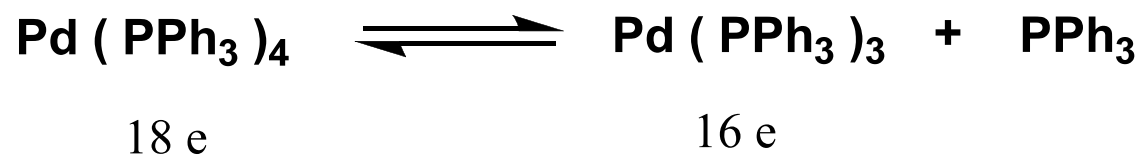


Metal complex in catalysis:

Large K value: Metal complex is too stable, Low catalytic activity

Small K value: Metal complex is too unstable, cannot be isolated

An appropriate K value is important for metal complex catalyst !



Chapter 5. Ligand Substitution Reactions

Outline

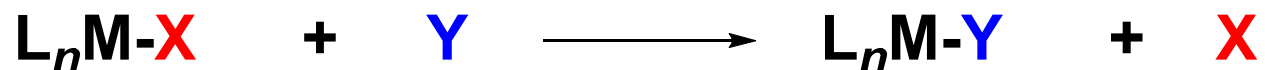
1. Introduction to ligand substitution
2. Dissociative substitution (解离型)
3. Associative substitution (缔合型)

References and suggested readings:

1. *The Organometallic Chemistry of the Transition Metals*, Robert H. Crabtree, 6th Edition, **2014**, Chapter 4.

1. Introduction to substitution reactions

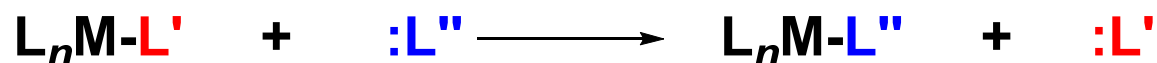
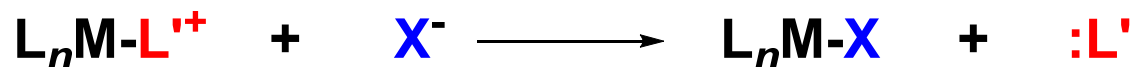
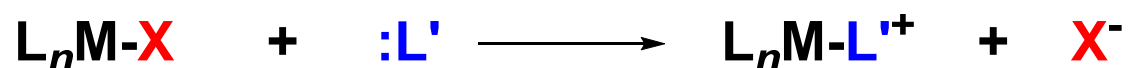
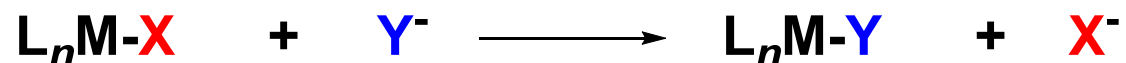
Basic reactions



Ligand Y is replacing ligand X

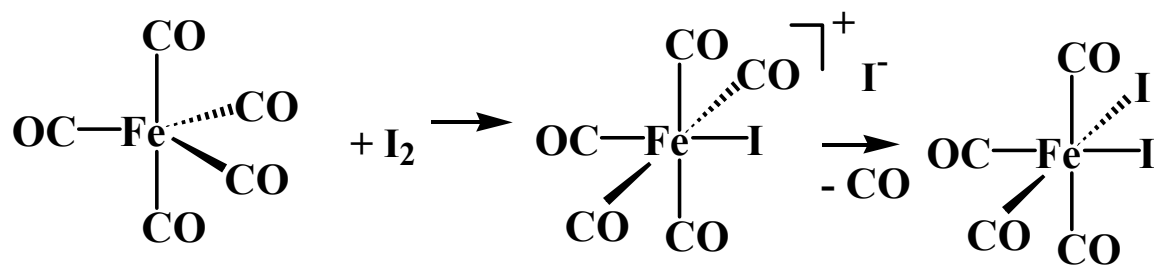
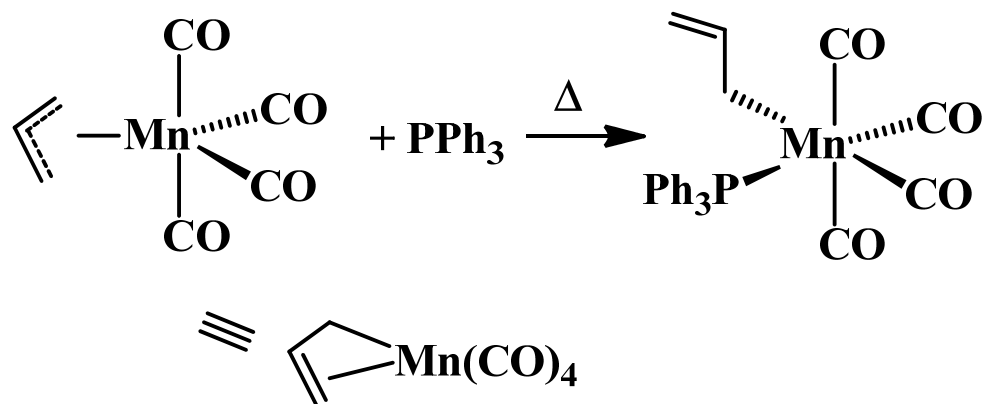
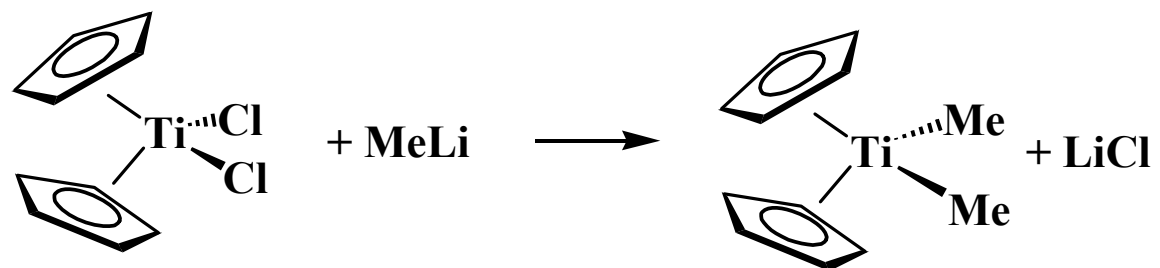
(Sometimes also referred as ligand exchange in the literature)

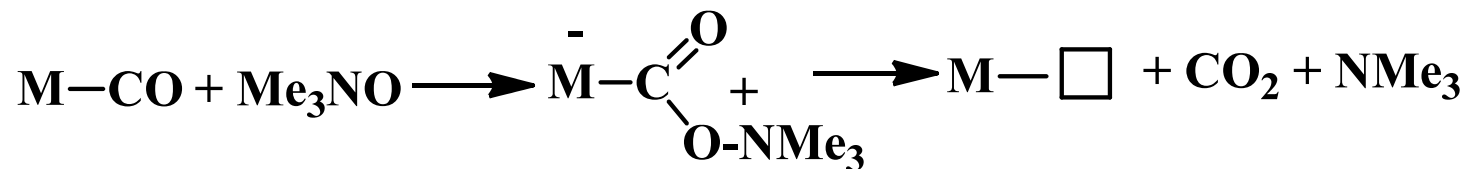
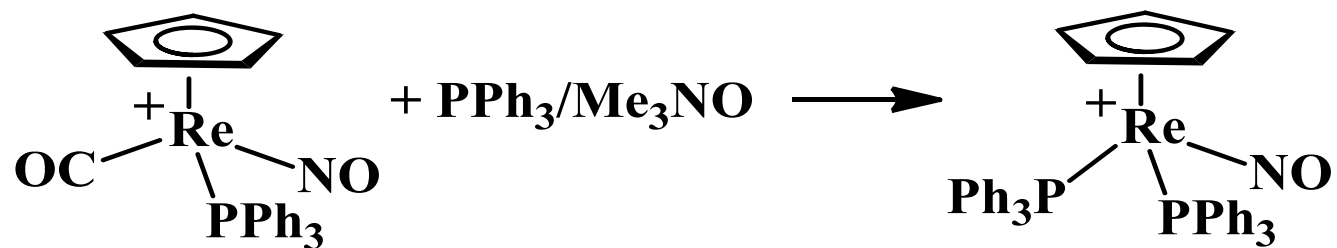
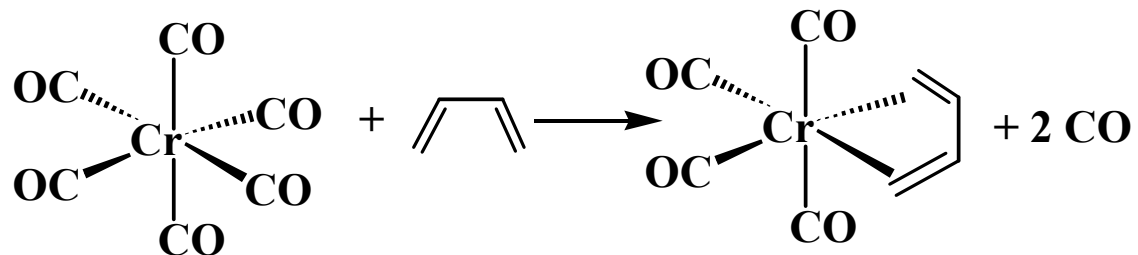
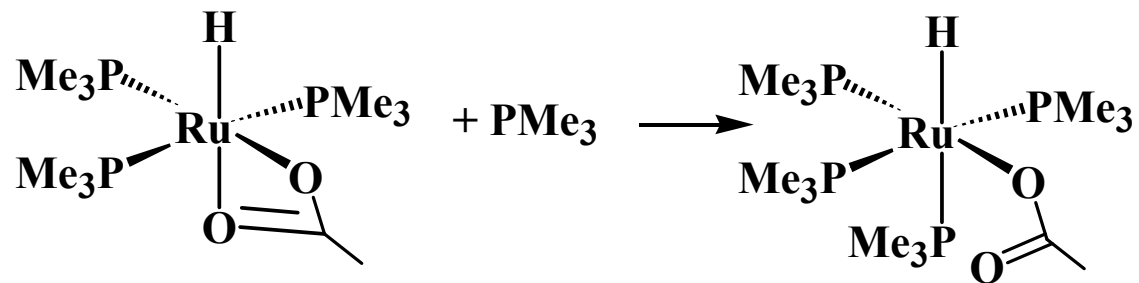
Types of substitution reactions:



2e $\rightarrow$ 2e, There is no change on the oxidation state of the involved M

Examples:





Major mechanisms for substitution reactions?



* **Dissociative substitution (D mechanism)** (解离取代反应)



rate determining step



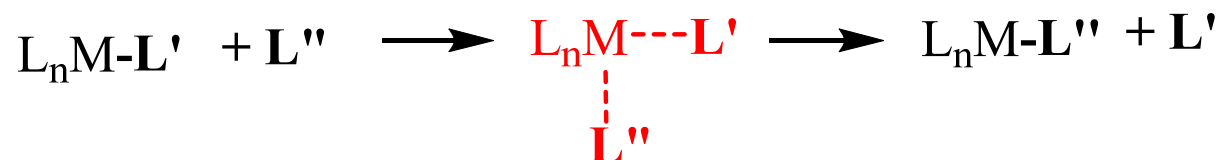
* **Associative substitution (A mechanism)** (缔合取代反应)



rate determining step



* **Interchange mechanism** (互换机理、交换取代)



Notes: Associative substitution goes through an intermediate while interchange mechanism goes through a transition state.

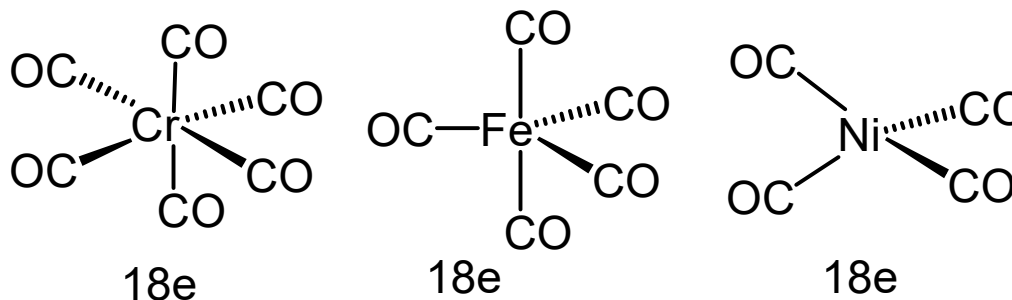
2. Dissociative substitution (解离取代反应)

Q1. What is the most important feature for D mechanism?



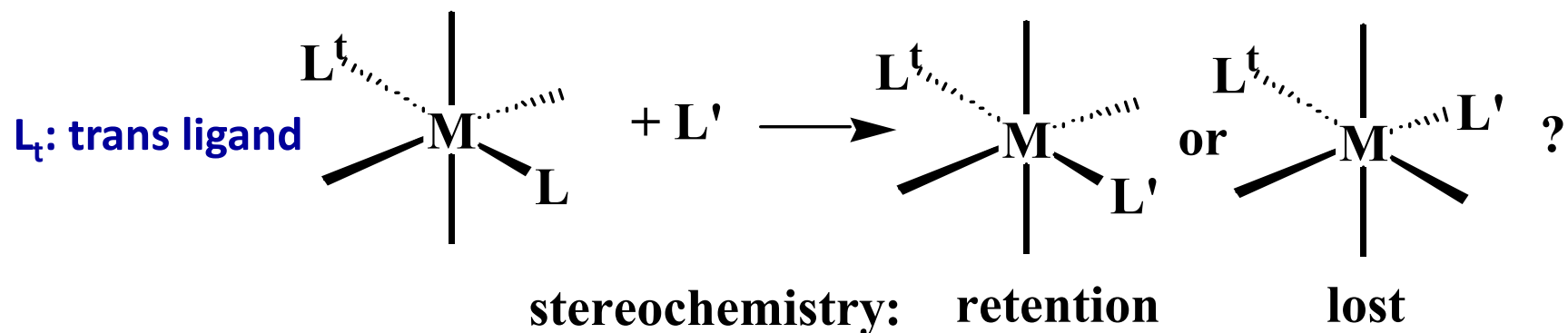
- * Rate independent of L''
- * Rate depends on $M-L'$ bond strength
- * $\Delta S^\ddagger > 0$
- * $\text{rate} = k[L_nM-L']$

Q2. What types of complexes are likely to undergo substitution reactions via D mechanism?

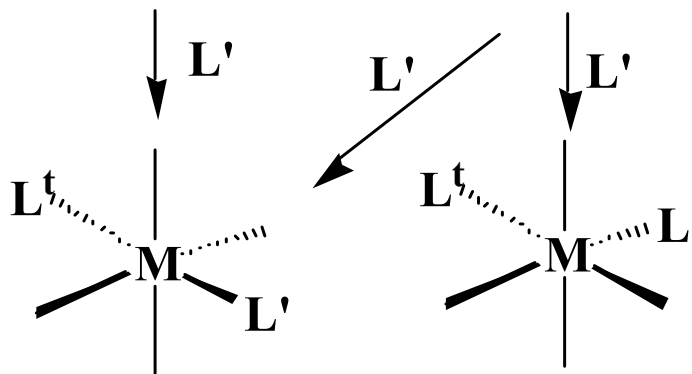
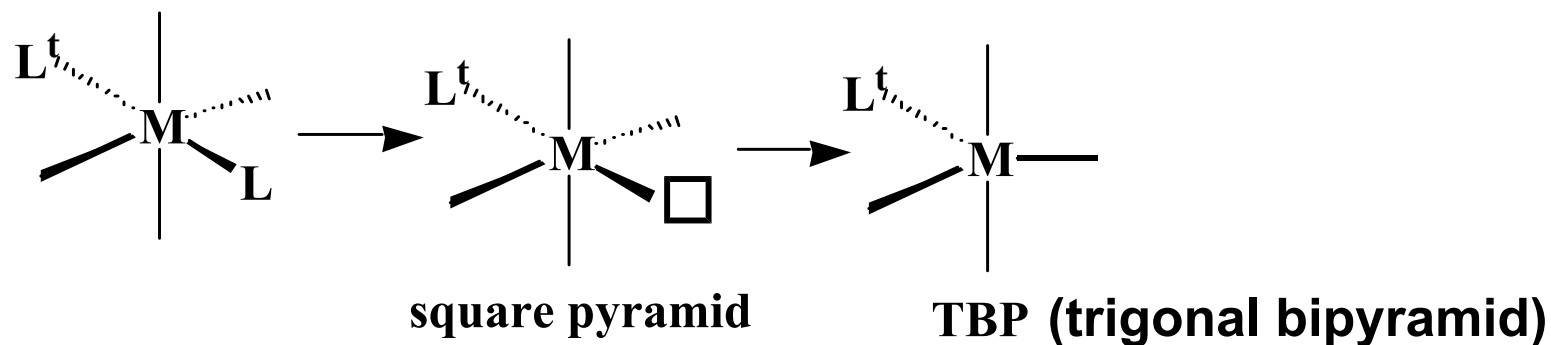


18e or > 18e species normally undergo D mechanism.

Q3. Stereochemistry retained or lost?

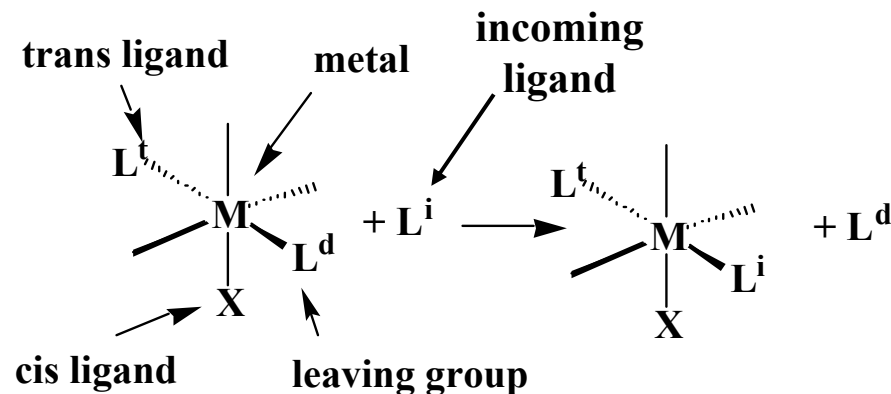


***depends on reactants**

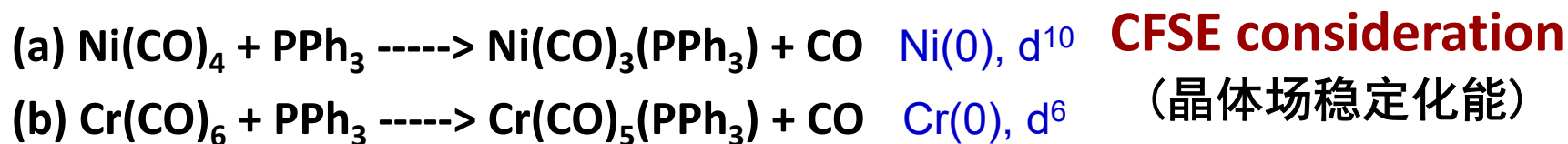


If L_t is a strong trans influence ligand, the square pyramid is stabilized, geometry is retained.

Q4. What affect reaction rates?



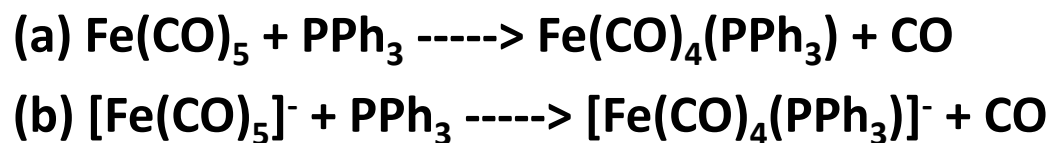
Q4A. Effect of metal. Consider the reactions below, which one is fast?



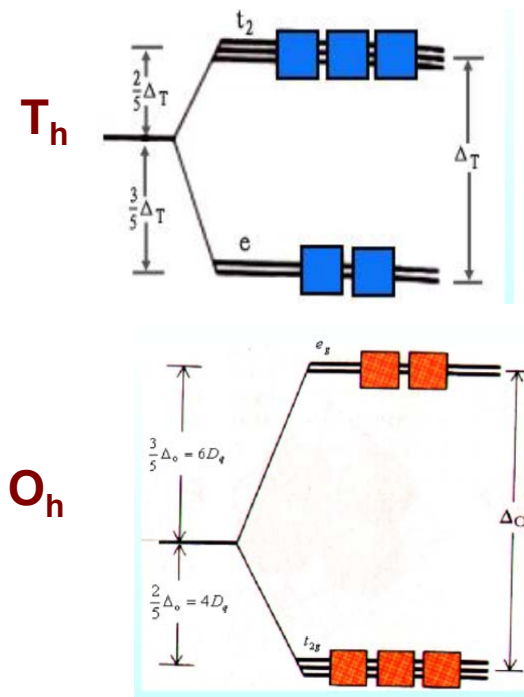
(a). Ni(0), d^{10} , no CFSE. Ni-CO bond is weaker.
 Cr(0), d^6 , big CFSE. Cr-CO bond is strong

** The substitution reactions of d^3 , d^6 (low spin) octahedral complexes are usually very slow.*

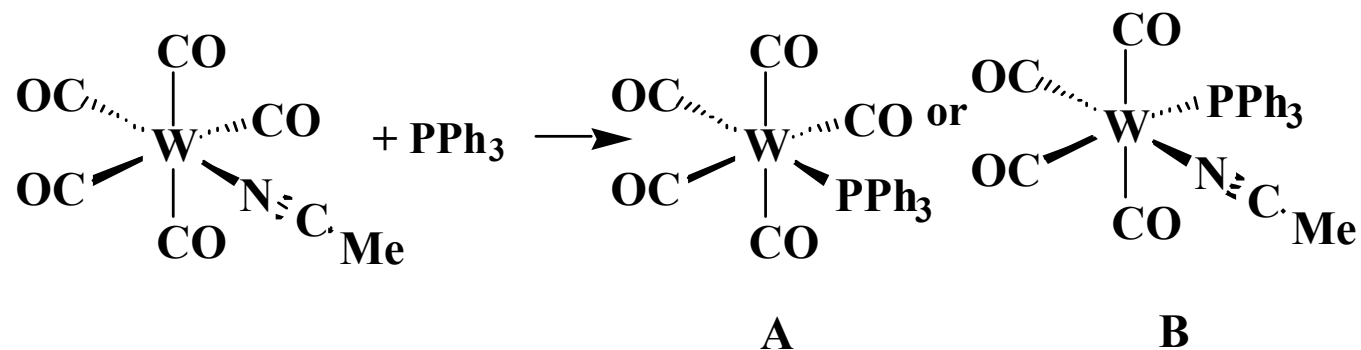
Q4B. Consider the reactions below, which one is fast?



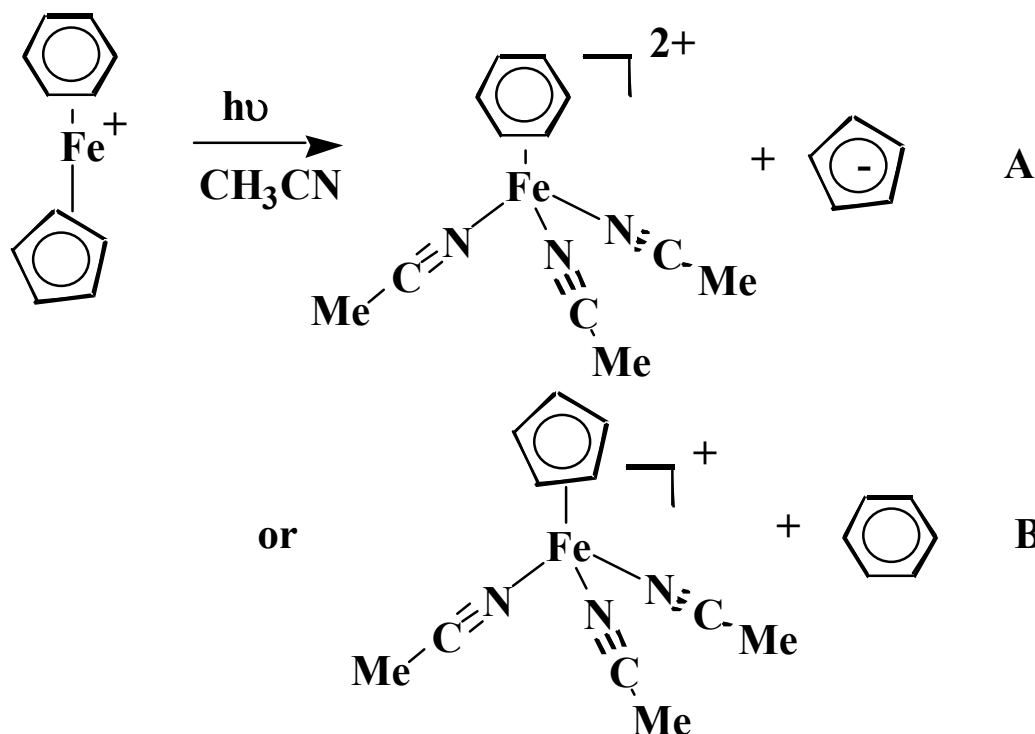
(b). $[\text{Fe}(\text{CO})_5]^-$ is a $19e^-$ complex. Fe-CO bond is weaker.



Q4C. Effect of leaving ligand. What is the likely product?



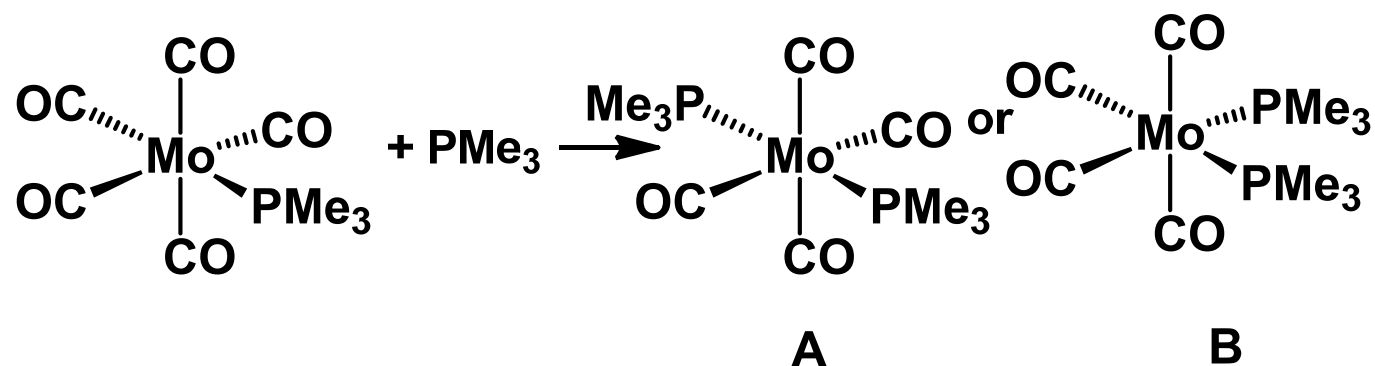
A. W-NCMe bond is weaker.



B. The more stable group is usually a better leaving group.

** Also chelating ligand dissociate less readily than monodentate ligands.*

Q4D. Effect of trans ligand. What is the likely product?



B. Because CO is a stronger trans influence ligand. B is also favored in terms of bonding.

- If bulky PR_3 is used instead of PMe_3 , A may be favored due to steric effect.*

Q4E. Effect of spectator ligand (steric effect). Consider the reactions below. Which one is fastest?

- (a) $\text{Ni(PPh}_3)_4 + \text{CO} \longrightarrow \text{Ni(CO)(PPh}_3)_3 + \text{PPh}_3$
- (b) $\text{Ni(PMe}_3)_4 + \text{CO} \longrightarrow \text{Ni(CO)(PMe}_3)_3 + \text{PMe}_3$
- (c) $\text{Ni(CO)(PMe}_3)_3 + \text{CO} \longrightarrow \text{Ni(CO)}_2(\text{PMe}_3)_2 + \text{PMe}_3$

(a). Size of PPh_3 is bigger. Thus PPh_3 will dissociate more readily.

3. Associative substitution

Q1. What is the most important feature for A mechanism?



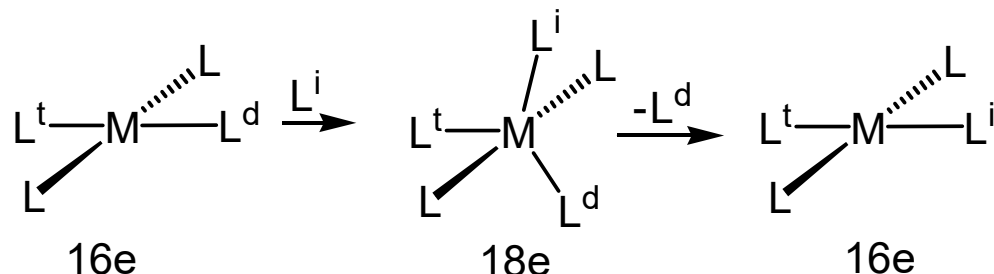
* Rate depends on both L_nM-L' and L''

$$\text{rate} = K[L_nM-L'][L'']$$

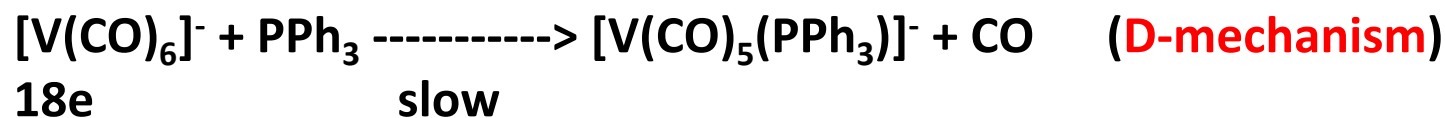
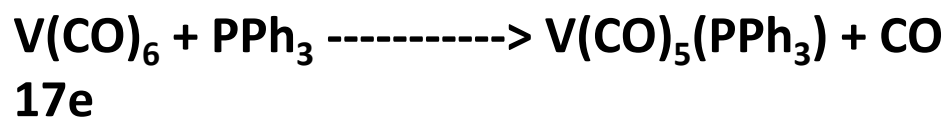
* $\Delta S^\ddagger < 0$

Q2. What types of complexes are likely to undergo substitution reactions via A mechanism?

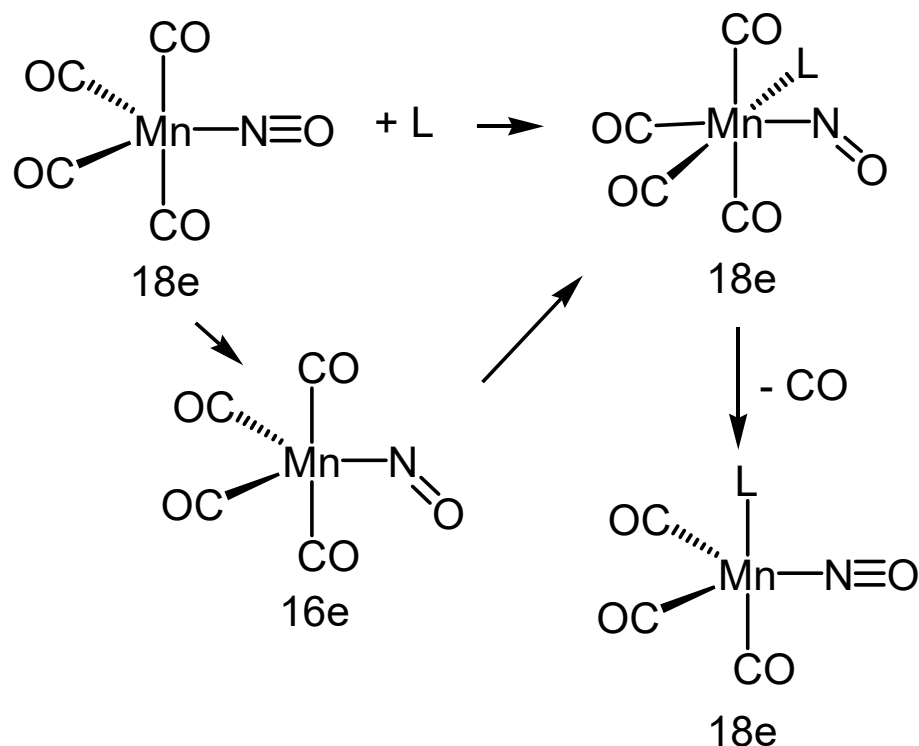
* *16e species normally undergo A mechanism.*



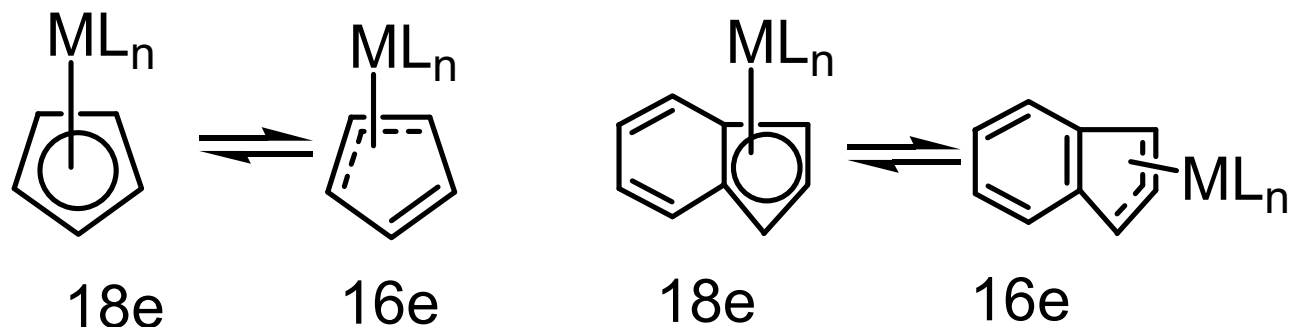
* *17e complexes. e.g.*



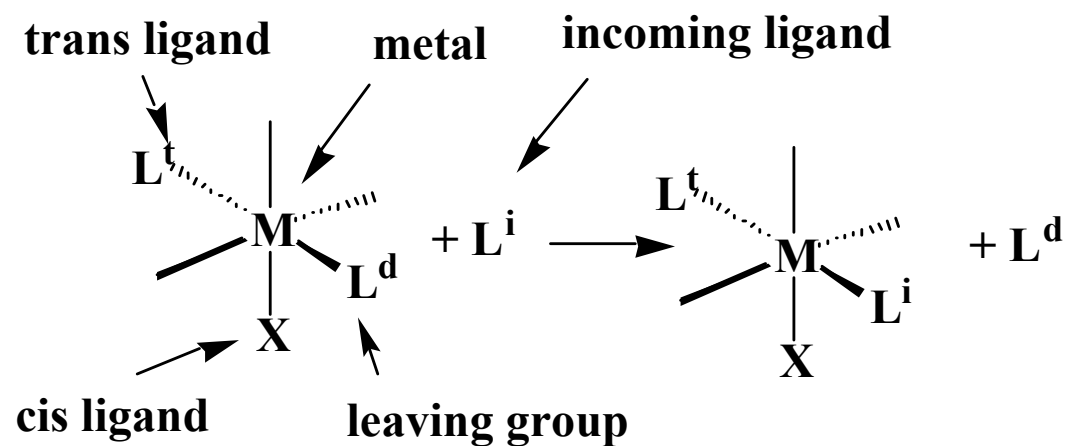
* *18e complexes that can rearrange to 16e species. e.g.*



* Other complexes capable of such rearrangement:



Q3. What affect reaction rates?



Practice Problems

1. Consider the following associative substitution reactions, which one reacts faster, PMe_3 or PPh_3 ?



PMe_3 . It is a better σ donor, V-P bond is stronger.

Which one reacts faster, PCy_3 or PPh_3 ?

PPh_3 . Although PCy_3 is a stronger σ donor, but is too bulky to approach to V.

2. 19e or 17e usually undergo substitution reactions fast than 18e complexes. Suggest a reason.

19e species, with an electron in a M–L σ^* orbital, are much more dissociatively labile than their 18e counterparts.

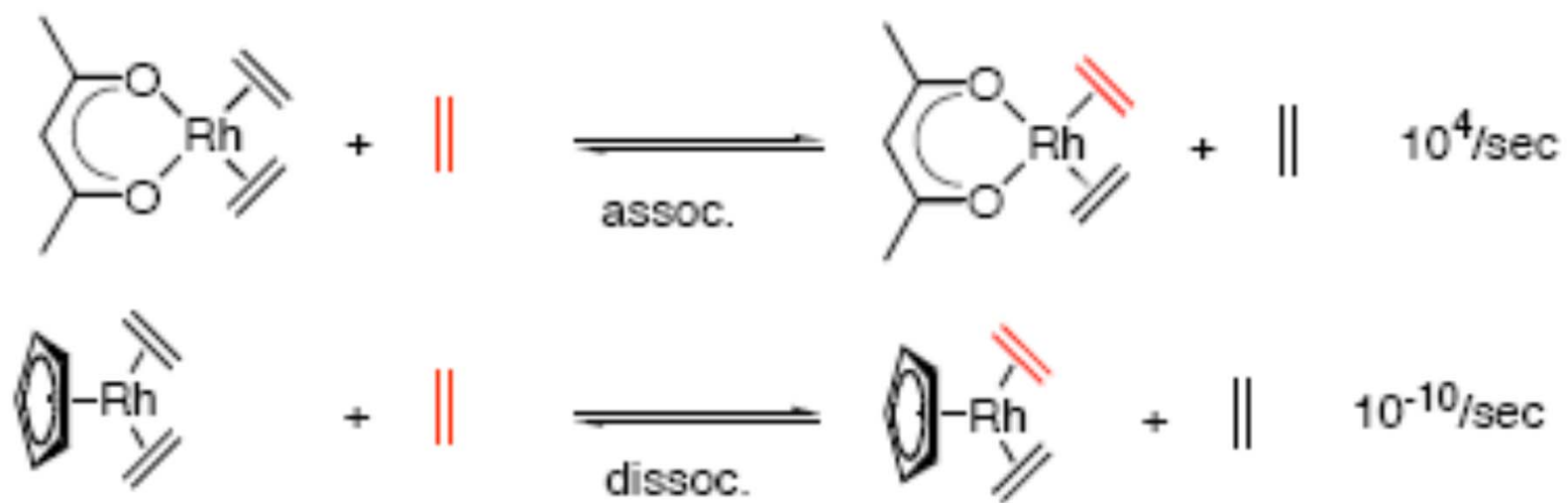
19e complexes, $\Rightarrow$ likely D mechanism

dissociate L is easier than 18e complexes.

17e complexes $\Rightarrow$ likely A mechanism

Add L is easier than 18e complexes.

Example:



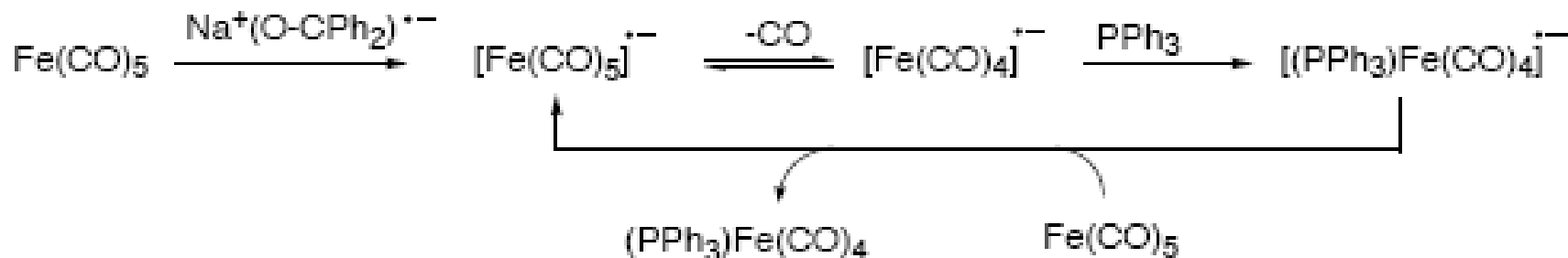
4. Redox effects

Two ways to make a coordination inert 18e complex give fast substitution are oxidation or reduction to a coordination labile 17e or 19e intermediate.

Catalysis by reduction

Reduction to 19e species can accelerate ligand substitution. This is commonly used for metal carbonyls.

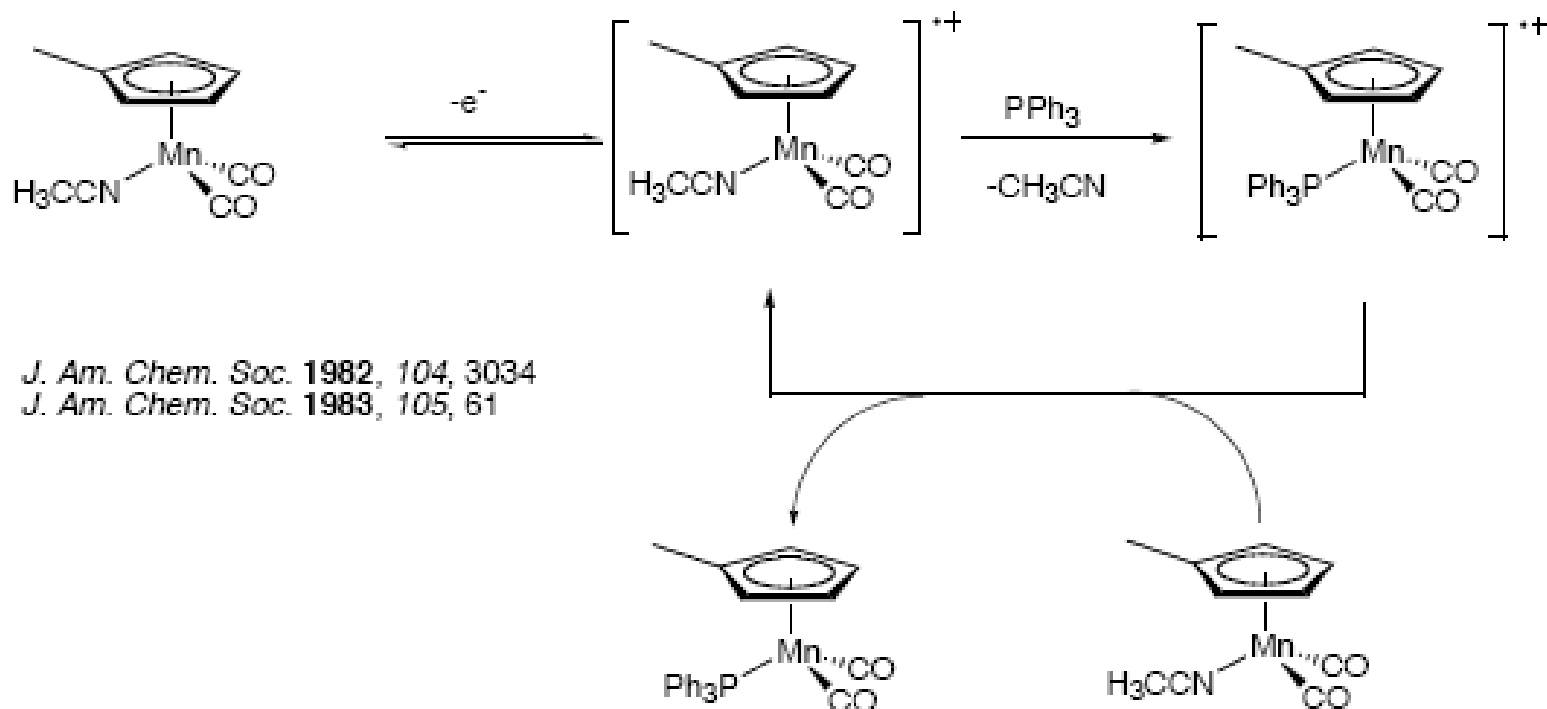
e.g. $\text{Fe}(\text{CO})_5$ can be substituted with electrochemical catalysis by a D-mechanism, where 19e $[\text{Fe}(\text{CO})_5]^{\cdot-}$ is the chain carrier in the catalytic cycle.



The initial product radical reduces the starting $\text{Fe}(\text{CO})_5$ so the cycle can continue.

Catalysis by oxidation

Oxidation of 18 electron complexes will give substitutionally labile 17 e complexes. Rapid ligand substitution can occur following the oxidation. Thus substitution reactions can be catalyzed by oxidant or electrochemically.



For this reaction to be catalytic, it is necessary that the reduction potential of the product should be more positive than that of the starting complex. In the case above the reduction potential of the starting complex is 0.15 V (vs. ferrocene/ferrocenium) and the E° of the product is 0.52 V.